

Syntax::tsntxi : \$Assumptions = b > 0, a > 0; is incomplete; more input is needed. “\ < \ >”

\$Assumptions = b > 0 & a > 0;

$$\phi = -\frac{GM}{\sqrt{R^2 + (a + \sqrt{z^2 + b^2})^2}}$$

$$-\frac{GM}{\sqrt{R^2 + (a + \sqrt{b^2 + z^2})^2}}$$

$$\text{d2}\phi\text{dR2} = D[D[\phi, R], R]$$

$$-\frac{3GMR^2}{\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{5/2}} + \frac{GM}{\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{3/2}}$$

FullSimplify[Together[d2φdR2]]

$$\frac{GM(a^2 + b^2 - 2R^2 + z^2 + 2a\sqrt{b^2 + z^2})}{\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{5/2}}$$

$$\text{d2}\phi\text{dz2} = D[D[\phi, z], z]$$

$$-\frac{3GMz^2(a + \sqrt{b^2 + z^2})^2}{(b^2 + z^2)\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{5/2}} + \frac{GMz^2}{(b^2 + z^2)\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{3/2}} - \frac{GMz^2(a + \sqrt{b^2 + z^2})}{(b^2 + z^2)^{3/2}\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{3/2}} +$$

$$\frac{GM(a + \sqrt{b^2 + z^2})}{\sqrt{b^2 + z^2}\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{3/2}}$$

Together[d2φdz2]

$$-\frac{1}{(b^2 + z^2)^{3/2}\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{5/2}}GM(-a^3b^2 - 3ab^4 - ab^2R^2 + ab^2z^2 + 4az^4 - 3a^2b^2\sqrt{b^2 + z^2} - b^4\sqrt{b^2 + z^2} - b^2)$$

$$\text{d}\phi\text{dR} = D[\phi, R]$$

$$\frac{GMR}{\left(R^2 + (a + \sqrt{b^2 + z^2})^2\right)^{3/2}}$$

\$Assumptions b > 0;

κ2 = FullSimplify[Together[(d2φdR2 + 3dφdR/R)/.{z->0}]]

$$\frac{GM(4(a+b)^2 + R^2)}{((a+b)^2 + R^2)^{5/2}}$$

ν2 = Simplify[d2φdz2/.{z->0}]

$$\frac{(a+b)GM}{b(a^2 + 2ab + b^2 + R^2)^{3/2}}$$

ν2/.{b->0.1a}

$$\frac{11.GM}{(1.21a^2+R^2)^{3/2}}$$

$$\phi h = \frac{-GM_h}{r+\alpha}$$

$$-\frac{GM_h}{r+\alpha}$$

$$\text{\$Assumptions} = R > 0;$$

$$\phi h_{cil} = \phi h /. \{r \rightarrow \sqrt{R^2 + z^2}\}$$

$$-\frac{GM_h}{\sqrt{R^2+z^2}+\alpha}$$

$$d2\phi h dR2 = D[D[\phi h_{cil}, R], R]$$

$$-\frac{2GR^2M_h}{(R^2+z^2)(\sqrt{R^2+z^2}+\alpha)^3} - \frac{GR^2M_h}{(R^2+z^2)^{3/2}(\sqrt{R^2+z^2}+\alpha)^2} + \frac{GM_h}{\sqrt{R^2+z^2}(\sqrt{R^2+z^2}+\alpha)^2}$$

$$\text{Simplify}[d2\phi h dR2 /. \{z \rightarrow 0\}]$$

$$-\frac{2GM_h}{(R+\alpha)^3}$$

$$d2\phi h dz2 = D[D[\phi h_{cil}, z], z]$$

$$-\frac{2Gz^2M_h}{(R^2+z^2)(\sqrt{R^2+z^2}+\alpha)^3} - \frac{Gz^2M_h}{(R^2+z^2)^{3/2}(\sqrt{R^2+z^2}+\alpha)^2} + \frac{GM_h}{\sqrt{R^2+z^2}(\sqrt{R^2+z^2}+\alpha)^2}$$

$$\text{Simplify}[d2\phi h dz2 /. \{z \rightarrow 0\}]$$

$$\frac{GM_h}{R(R+\alpha)^2}$$

$$\text{\$Assumptions} = b > 0;$$

$$v_c = \sqrt{R \text{Simplify}[D[\phi, R] /. \{z \rightarrow 0\}] + r D[\phi h, r]}$$

$$\sqrt{\frac{GMR^2}{(a^2+2ab+b^2+R^2)^{3/2}} + \frac{GrM_h}{(r+\alpha)^2}}$$

$$d\phi h dR = D[\phi h_{cil}, R]$$

$$\frac{GRM_h}{\sqrt{R^2+z^2}(\sqrt{R^2+z^2}+\alpha)^2}$$

$$\text{\$Assumptions} = R > 0;$$

$$\kappa 2 h = \text{Simplify}[(d^2 \phi h d R^2 + 3 d \phi h d R / R) /. \{z \rightarrow 0\}]$$

$$\frac{G(R+3\alpha)M_h}{R(R+\alpha)^3}$$

$$\kappa 2 + \kappa 2 h$$

$$\frac{GM(4a^2+8ab+4b^2+R^2)}{(a^2+2ab+b^2+R^2)^{5/2}} + \frac{G(R+3\alpha)M_h}{R(R+\alpha)^3}$$

$$\text{\$Assumptions} = b > 0;$$

$$\text{FullSimplify}[d \phi d R /. \{z \rightarrow 0\}]$$

$$\frac{GMR}{((a+b)^2+R^2)^{3/2}}$$

$$\text{\$Assumptions} = R > 0;$$

$$\text{FullSimplify}[d \phi h d R /. \{z \rightarrow 0\}]$$

$$\frac{GM_h}{(R+\alpha)^2}$$

$$\text{\$Assumptions} = b > 0 \&\& R > 0;$$

$$\text{FullSimplify}[R d \phi d R + R d \phi h d R /. \{z \rightarrow 0\}]$$

$$GR \left(\frac{MR}{((a+b)^2+R^2)^{3/2}} + \frac{M_h}{(R+\alpha)^2} \right)$$